EX.3 Time series

1. Verify that the autoregressive process

$$X_t = (X_{t-1} + X_{t-2})/12 + Z_t$$

where $\{Z_t\}$ is white noise , is stationary, and find its autocorrelation function.

2. Plot the autocorrelation function (up to lag 10) of the second order autoregressive process

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + Z_t,$$

where $\{Z_t\}$ is white noise, for each of

$(\alpha_1 , \alpha_2) = (0.3,0.5)$ and $(0.3,-0.5)$.

3. Consider the process

$$X_t = A\cos(wt) + B\sin(wt),\ t = 0,1,2,...,$$

where A and B are uncorrelated random variables with zero mean and unit variance, w is a constant and $w \in [0, \pi]$. Show that X_t is stationary.

4. You are given the ARMA process:

$$y_t = 12 + 0.4y_{t-1} + 0.1y_{t-2} + \varepsilon_t + 0.3\varepsilon_{t-1} - 0.2\varepsilon_{t-2}$$

Find the mean of y_t.

5. Find the autocorrelation function for the process defined by $Z_t = 5 + a_t - 0.5a_{t-1} + 0.25a_{t-2}$, where $\{a_t\}$ is a white noise process.

6. For the ARMA(1,2) model

$$Z_t = 0.8Z_{t-1} + a_t + 0.7a_{t-1} + 0.6a_{t-2}$$

Show that (a) $\rho_k = 0.8\rho_{k-1}$ *for* $k \geq 3$ and

(b) $\rho_2 = 0.8\rho_1 + 0.6\sigma_a^2 / \gamma_0$.

Here $\gamma_0 = E(Z_t^2)$

7. For the process

$$y_t = 0.3y_{t-1} + 0.2y_{t-2} + \varepsilon_t - \theta_1\varepsilon_{t-1} - \theta_2\varepsilon_{t-2}$$

you are given $\rho_1 = 0.8,\ \rho_2 = -0.2.$ Find ρ_4.

Hint : Is y_t stationary?